# Graph Neural Networks

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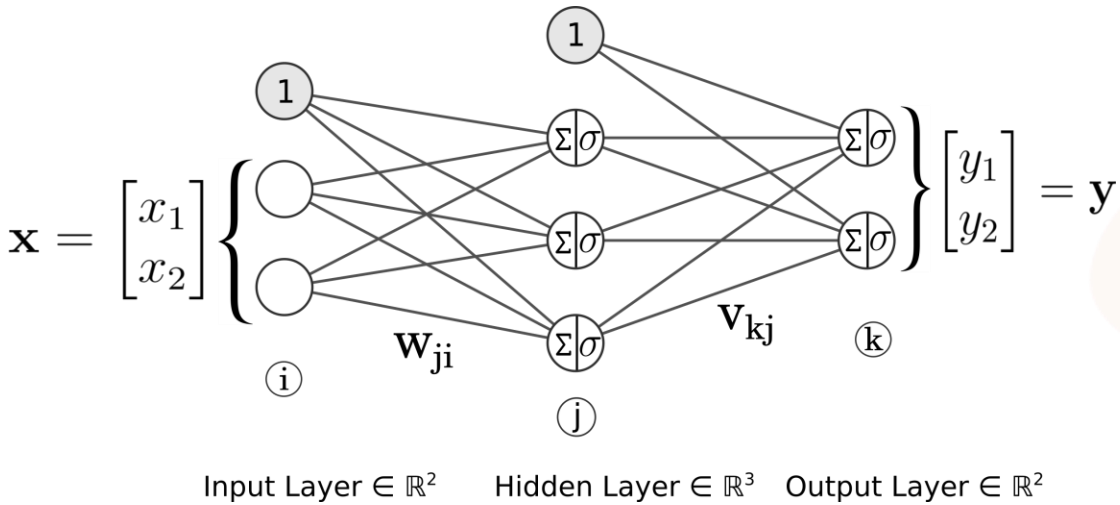
# Mathematics of Classical Deep Learning

# Forward Propagation

The process used to generate the output of a neural network




| Weight Initialization Techniques | Activation Function Match |
|---|---|
| Xavier/Glorot[1] | Tanh, Softmax |
| He | ReLU and Variants |
| LeCun | SeLU |
| Normal | --- |
| Zero | --- |

Keynote of this process:

- Weight initialization
- The mathematical computation

Mathematical computation of hidden layer

$$\mathbf{L} = \begin{bmatrix} 4 & 1 \\ 4.4 & 0.3 \\ 7.5 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 5.6 \\ 43.5 \end{bmatrix}$$

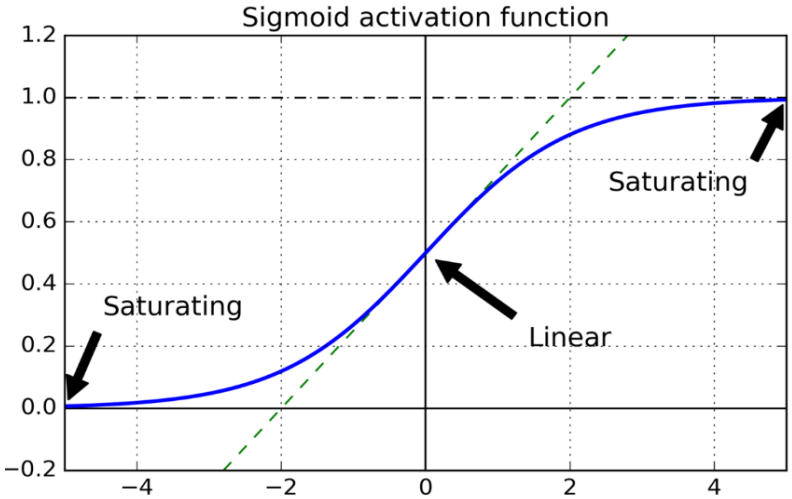
[1] Understanding the difficulty of training deep feedforward neural networks (Glorot and Bengio, 2010)

# Forward Propagation (FP)

## Vanishing and Exploding Gradient Problem

The problem where the gradient get smaller (vanishing) or bigger (exploding)

**Signal voice** ⇨ Amplifier → Amplifier → Amplifier → Amplifier ⇨ **Amplified signal**




**Var(input) >> Var(output)**

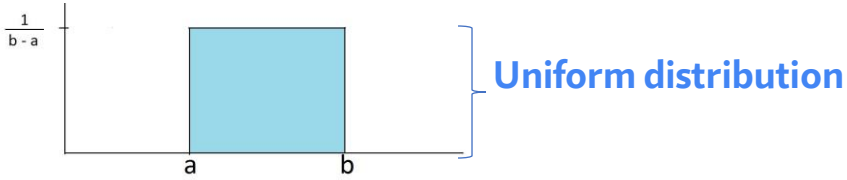
The imbalanced variance is the main problem !

All need to be set properly in order for your voice to come out loud and clear at the end of the chain. NNs are equal!!

**Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (Aurélien Géron, 2019)**

# Forward Propagation (FP)

## Xavier Uniform Initialization

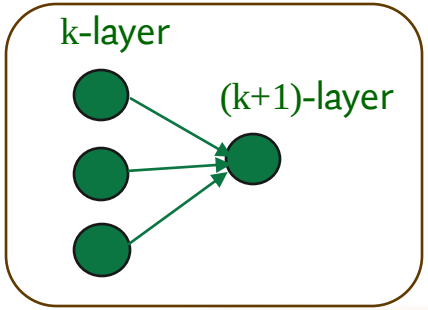



1. $$Var[w_i] = \frac{2}{n^k + n^{k+1}}$$

2. $$w_i \sim \mathcal{U}(-a, a) \rightarrow Var = \frac{1}{12}(a - (-a))^2$$

$$Var = \frac{a^2}{3}$$

3. $$a = \sqrt{\frac{6}{n^k + n^{k+1}}}$$




The variance of weight distribution is the key point to understanding the Xavier Initialization

**Algorithm 1** Xavier uniform weight initialization

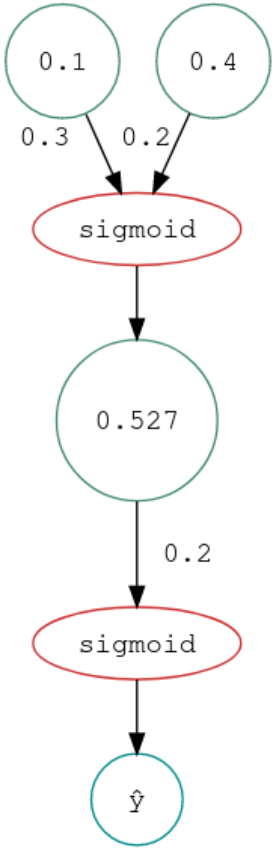
**Require:** $k \geq 2$

```
n_k ← number of neurons in k-layer
n_{k+1} ← number of neurons in (k+1)-layer
weights ← zero matrix initialization
for i = 0 to n_k do
    for j = 0 to n_{k-1} do
        weight ← U(-a, a)
        weights[i, j] ← weight
    end for
end for
```

# Backpropagation

**Dataset**

| x1 | x2 | y |
|---|---|---|
| 0.1 | 0.4 | 1 |




**1 Compute the output ŷ by the forward propagation**

$$h = \sum_i w_i x_i = 0.1 \times 0.3 + 0.4 \times 0.2$$

$$h = \sigma(h) = \texttt{sigmoid}(0.11)$$

$$h = \frac{e^{0.11}}{e^{0.11} + 1}$$

$$h = 0.527$$

$$\hat{y} = \sigma(\mathbf{w}_o h)$$

$$\hat{y} = \texttt{sigmoid}(0.2 \times 0.527)$$

$$\hat{y} = \frac{e^{0.105}}{e^{0.105} + 1}$$

$$\hat{y} = 0.526$$

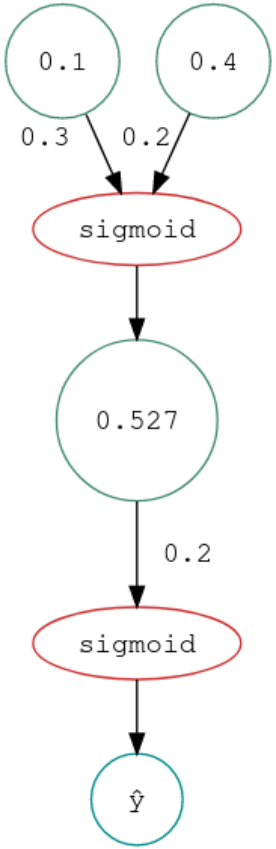
# Backpropagation

**Objective Function**

$$e(\mathbf{w}, x, y) = (y - \hat{y})\sigma(\mathbf{w}x)$$

**Sigmoid derivative**

$$\sigma'(x) = \texttt{sigmoid}'(x)$$
$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$




2. **Start the backward pass computing the error term**

$$\frac{\partial e(\mathbf{w}_o, x, y)}{\partial \mathbf{w}_o} = \delta^o = (y - \hat{y})\sigma'(\mathbf{w}_o h)$$

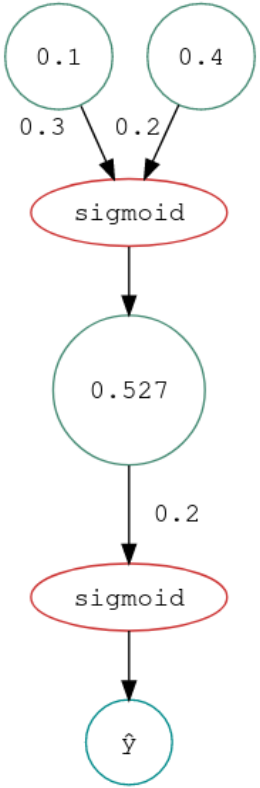
- **For the output weight**

$$\delta^o = (1 - 0.526)(0.526)(1 - 0.526)$$
$$\delta^o = 0.118$$

- **For the hidden unit weight**

$$\delta^h = \mathbf{w}_o \delta^o \sigma'(h)$$
$$\delta^h = 0.2 \times 0.118 \times (0.527)(1 - 0.527)$$
$$\delta^h = 0.0058$$

# Backpropagation




**❸ Update all the weights (Delta rule) "See also Hebb Rule [2]"**

$$\Delta w = \eta(\text{learning rate}) \times \delta(\text{error term}) \times a(\text{neuron input})$$

- **For the output weight** $(\eta = 0.5)$

$$\Delta w_o = 0.5 \times 0.118 \times 0.526$$

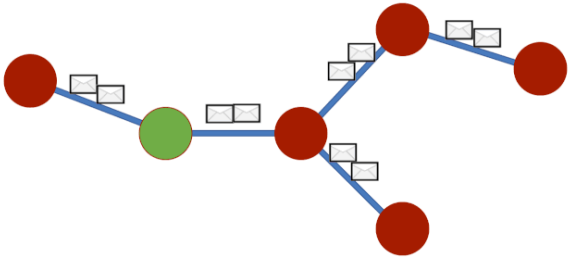
$$\Delta w_o = 0.031$$

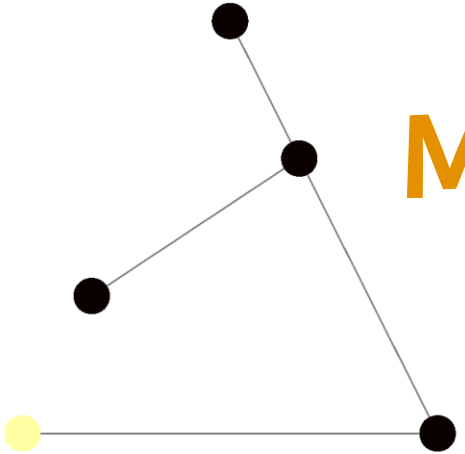
- **For the hidden unit weight** $(\eta = 0.5)$

$$\Delta w_1 = 0.5 \times 0.0058 \times 0.1 \qquad \Delta w_2 = 0.5 \times 0.0058 \times 0.4$$

$$\Delta w_1 = 0.00029 \qquad \Delta w_2 = 0.0011$$

[2] https://medium.com/@reutdayan1/hebbian-learning-biologically-plausible-alternative-to-backpropagation-6ee0a24deb00





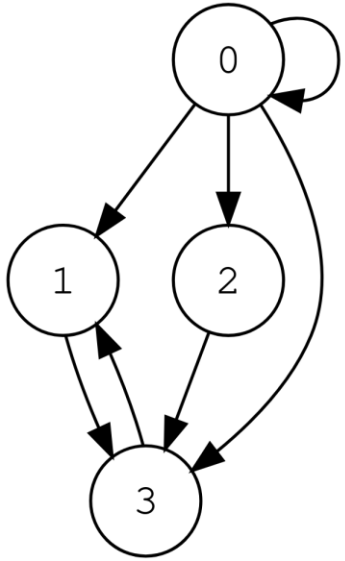
# Message Passing on Graphs

"In the simplest form, is just the multiplication between an adjacency matrix and the nodes features of a graph"

# Message Passing on Graphs

The key concept is propagating the information from one node to another using the adjacency matrix of the graph

## Graph Representation




1

## Graph Adjacency Matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

## Node Features

$$\mathbf{X} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

2

## Simple Message Passing

$$\overbrace{\mathbf{H} = \mathbf{A}\mathbf{X}}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

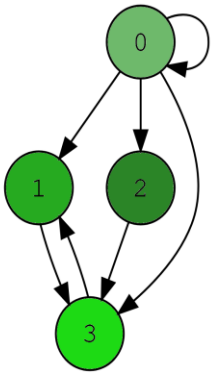
# Message Passing on Graphs

## Computation

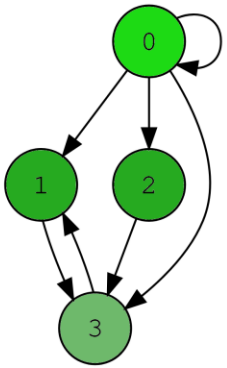
Node Features

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \times 1 + 1 \times 2 + 1 \times 3 + 1 \times 4 \\ 0 \times 1 + 0 \times 2 + 0 \times 3 + 1 \times 4 \\ 0 \times 1 + 0 \times 2 + 0 \times 3 + 1 \times 4 \\ 0 \times 1 + 1 \times 2 + 0 \times 3 + 0 \times 4 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \\ 4 \\ 2 \end{bmatrix}$$

Graph Adjacency Matrix




After Message Propagation

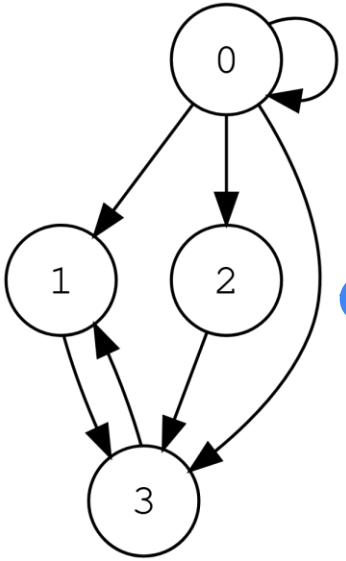



Aggregated Features

# Message Passing on Graphs

The key concept is propagating the information from one node to another using the adjacency of the graph

## Graph Representation




1

### Graph Adjacency Matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

### Node Features

$$\mathbf{X} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

### Inverse of Degree Matrix

$$\mathbf{D}^{-1} = \begin{bmatrix} 0.25 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2

### Average Message Passing

$$\overbrace{\mathrm{H} = \mathrm{D}^{-1}\mathrm{AX}}$$

$$\mathbf{H} = \begin{bmatrix} 0.25 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

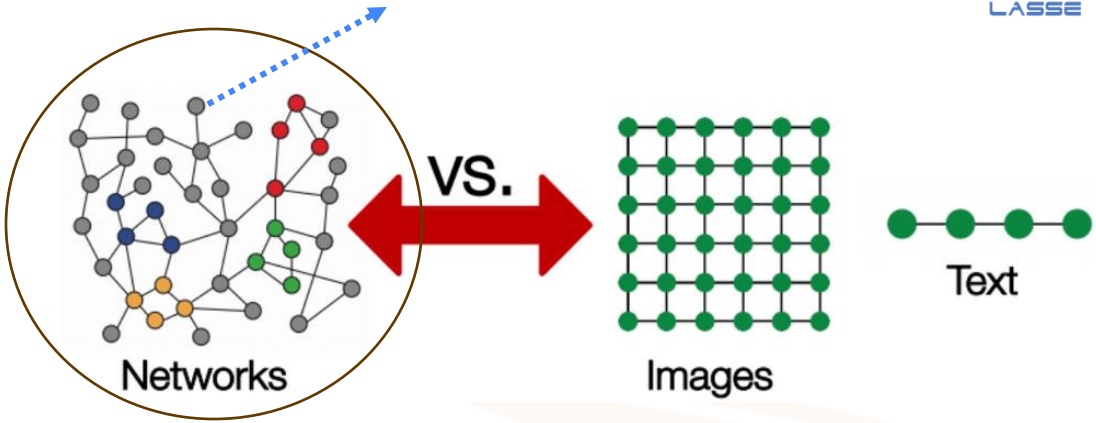
# Graph Neural Networks (GNN)

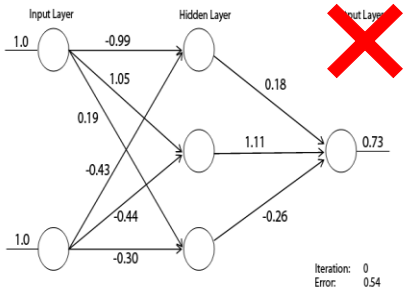
"Graphs are ubiquitous in our world, network communication, social networks, chemistry molecules … well, GNN can processed it"

# Graph Neural Networks

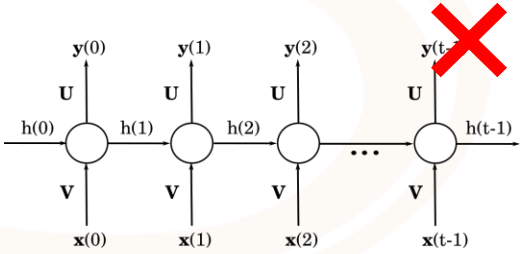
Non-euclidean data

**Are the classical ANN models capable of handling graphs (non-Euclidean) data?**



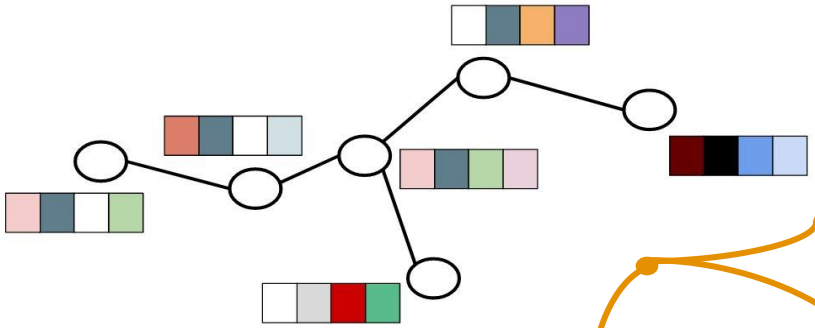

**Individually, these models can't handle well the complexities inherent to the relation between the nodes and their connecting edges**

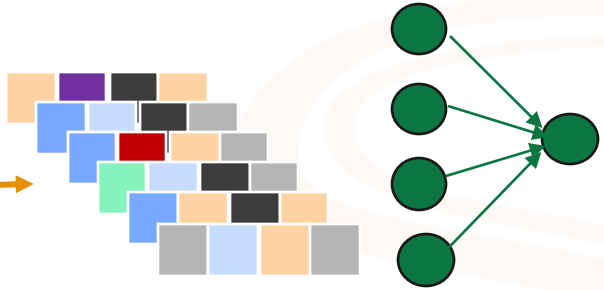


# Graph Neural Networks

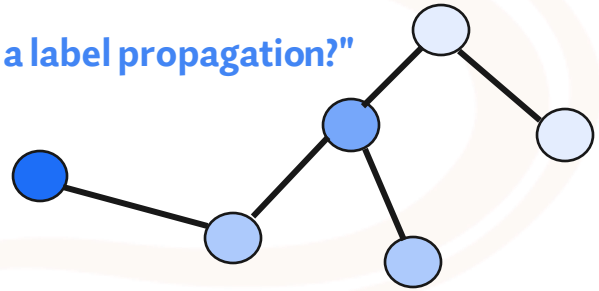
## How to process non-Euclidean data

"Use a simple classical neural network model ?"





"Perform a label propagation?"



"Use a graph neural network (GNN)?"

# Graph Convolutional Networks (GCN)

"The cornerstone graph neural network model"

# Graph Convolutional Neural Networks

## Mathematical Description

$$\mathbf{H}^{(l+1)} = f(\mathbf{H}^{(l)}, \mathbf{A})$$

Basic Neural Network Layer Definition

Message Passing

$$f(\mathbf{H}^{(l)}, \mathbf{A}) = \sigma(\mathbf{A}\mathbf{H}^{(l)}\mathbf{W})$$

Simplest GCN Definition

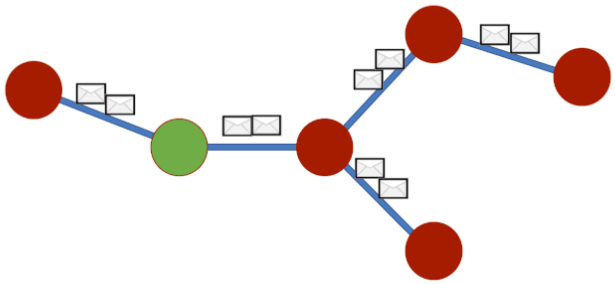
Where:

$\mathbf{H}^{(l)}$ – NN output of $l$-th layer

$\mathbf{A}$ – Adjacency Matrix

$\mathbf{W}$ – Weight Matrix

# Graph Convolutional Neural Network (GCN)



$$g_\theta * x = \mathbf{U} g_\theta \mathbf{U}^\top$$

Complexity way to perform convolution on graphs

Approximation

$$\mathrm{H} = \hat{\mathrm{D}}^{-\frac{1}{2}} \hat{\mathrm{A}} \hat{\mathrm{D}}^{-\frac{1}{2}} \mathrm{X}\mathrm{W}$$

$$\mathrm{A} + \mathrm{I}$$

Normalized Adjancency Matrix

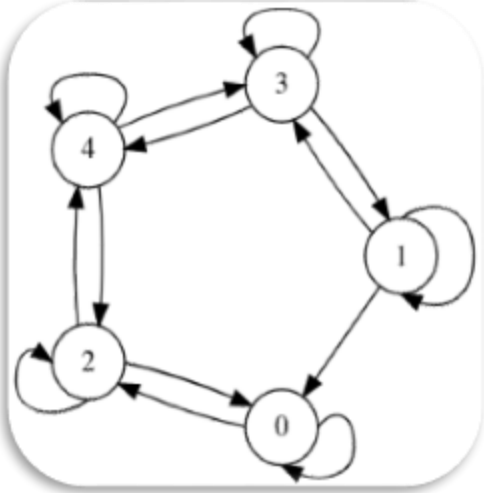
It is a GNN model aimed to use for classification tasks. It was created with a similar intuition as Convolutional Neural Networks (CNN)

The starting point comes from a spectral graph convolution, yet introduces some simplifications

# Graph Convolutional Neural Networks

## Modeling GCN FP with 1 Hidden Layer for Node Classification

Considering as input the below graph with 5 nodes




Which the adjacency matrix is given by

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

# Graph Convolutional Neural Networks

## Modeling GCN FP with 1 Hidden Layer for Node Classification

Each node has 4 features

It is also considered a weight matrix and implicit bias given by

$$\mathbf{H}^{(0)} = \mathbf{X} = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \quad \mathbf{W} = \overbrace{\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 1 \end{bmatrix}}^{\text{Weight Matrix}} \quad \mathbf{b} = \overbrace{\begin{bmatrix} 0 & 0 & -1 \end{bmatrix}}^{\text{Bias}}$$

# Graph Convolutional Neural Networks

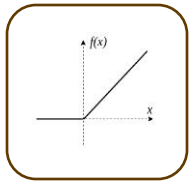
## Modeling GCN FP with 1 Hidden Layer for Node Classification

**Input Graph**

**Forward Propagation**

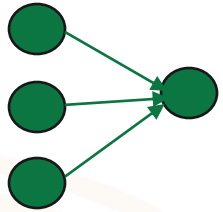
GCN Hidden Layer

**Non-linear Activation Function**




GCN Output

**Multilayer Perceptron**

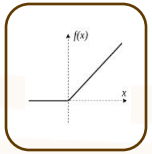


"For each node in the graph: classify"

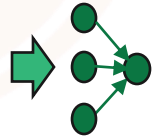
---

$$\mathbf{H}^{(0)} = \mathbf{X} = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \Rightarrow \underbrace{\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 1 \end{bmatrix}}_{\mathbf{AH}^{(0)}\mathbf{W}} + \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & -1 \\ 0 & 0 & -1 \\ 0 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix} \Rightarrow$$

**ReLU**



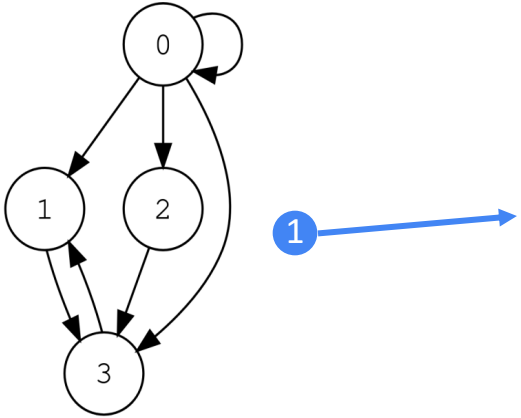

$$\Rightarrow \begin{bmatrix} 4 & 1 & 2 \\ 4 & 0 & 5 \\ 5 & 1 & 3 \\ 2 & 0 & 4 \\ 2 & 0 & 2 \end{bmatrix} \Rightarrow$$



# Message Passing Based on Neural Networks

"The use of neural network models to perform the well-known message passing"

# Message Passing Using an RNN




**Node Features**

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

**RNN Forward Propagation Law**

$$\mathbf{h}_t = \sigma(\mathbf{h}_{t-1}\mathbf{W} + \mathbf{x}_t\mathbf{V})$$

$$\mathbf{y}_t = \mathbf{h}_t\mathbf{U}$$

**Weights and hidden states**

$$\mathbf{W} = \begin{bmatrix} 1 & 4 \\ 8 & 2 \end{bmatrix} \quad \mathbf{V} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

$$\mathbf{U} = \begin{bmatrix} -1 & 1 \end{bmatrix}$$

$$\mathbf{h}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

**In t=1 we have, considering a ReLU activation function:**

$$\mathbf{h}_1 = \mathtt{ReLU}\left(\begin{bmatrix} 1 & 4 \\ 8 & 2 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \end{bmatrix} + 1 \times \begin{bmatrix} 6 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

$$\mathbf{y}_1 = \begin{bmatrix} 6 \\ 4 \end{bmatrix}\begin{bmatrix} -1 & 1 \end{bmatrix} = -2$$

# Message Passing Using RNNs

## All Computations

### At t=1:

$$\mathbf{h}_1 = \mathtt{ReLU}\left(\begin{bmatrix}1 & 4\\ 8 & 2\end{bmatrix}\begin{bmatrix}0\\ 0\end{bmatrix} + 1 \times \begin{bmatrix}6\\ 4\end{bmatrix}\right) = \begin{bmatrix}6\\ 4\end{bmatrix}$$

$$\mathbf{y}_1 = \begin{bmatrix}6\\ 4\end{bmatrix}\begin{bmatrix}-1 & 1\end{bmatrix} = -2$$

### At t=2:

$$\mathbf{h}_2 = \mathtt{ReLU}\left(\begin{bmatrix}1 & 4\\ 8 & 2\end{bmatrix}\begin{bmatrix}6\\ 4\end{bmatrix} + 2 \times \begin{bmatrix}6\\ 4\end{bmatrix}\right) = \begin{bmatrix}34\\ 64\end{bmatrix}$$

$$\mathbf{y}_2 = \begin{bmatrix}34\\ 64\end{bmatrix}\begin{bmatrix}-1 & 1\end{bmatrix} = 30$$

### At t=3:

$$\mathbf{h}_3 = \mathtt{ReLU}\left(\begin{bmatrix}1 & 4\\ 8 & 2\end{bmatrix}\begin{bmatrix}34\\ 64\end{bmatrix} + 3 \times \begin{bmatrix}6\\ 4\end{bmatrix}\right) = \begin{bmatrix}308\\ 412\end{bmatrix}$$

$$\mathbf{y}_3 = \begin{bmatrix}308\\ 412\end{bmatrix}\begin{bmatrix}-1 & 1\end{bmatrix} = 104$$

### At t=4:

$$\mathbf{h}_4 = \mathtt{ReLU}\left(\begin{bmatrix}1 & 4\\ 8 & 2\end{bmatrix}\begin{bmatrix}308\\ 412\end{bmatrix} + 4 \times \begin{bmatrix}6\\ 4\end{bmatrix}\right) = \begin{bmatrix}1980\\ 3304\end{bmatrix}$$

$$\mathbf{y}_4 = \begin{bmatrix}1980\\ 3304\end{bmatrix}\begin{bmatrix}-1 & 1\end{bmatrix} = 1324$$

### Hidden States

| | | | |
|---|---|---|---|
| 6 | 34 | 308 | 1980 |
| 4 | 64 | 412 | 3304 |

### Outputs

| | | | | |
|---|---|---|---|---|
| y | -2 | 30 | 104 | 1324 |

# Programming a GNN Model

"A Hands-on approach in how to use the main packages to create a GNN model"

# Programming a GNN Model

## Frameworks

Deep Graph Library

TensorFlow GNN 2.0

Pytorch Geometric

iGNNition

# An Application for Network Communication Modeling

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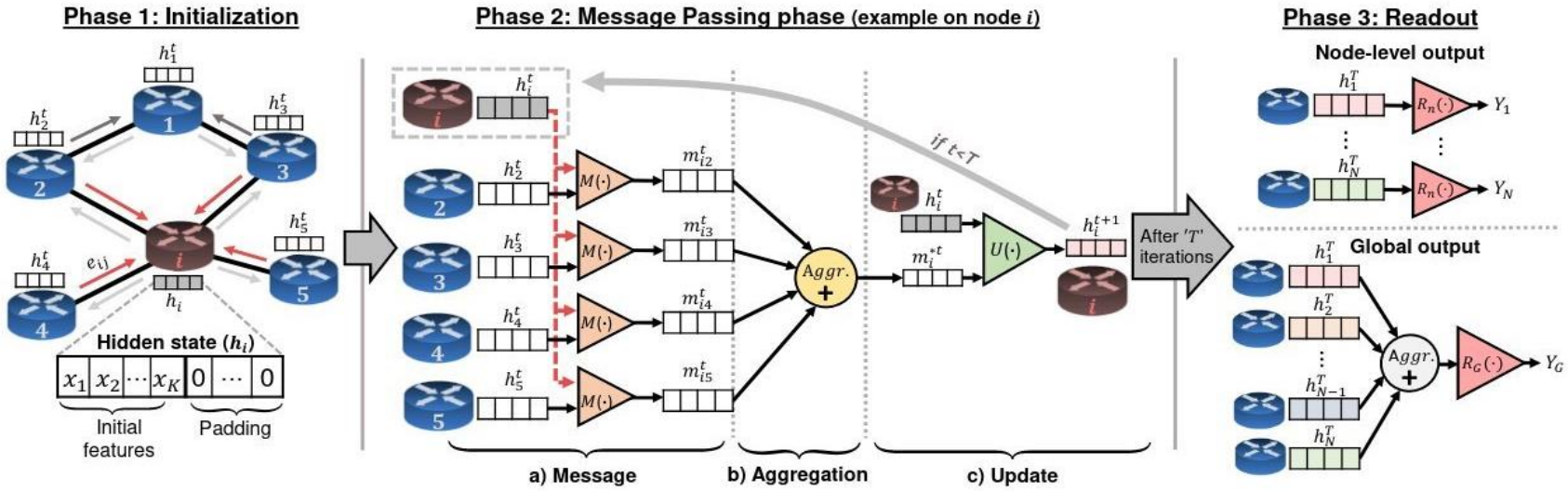
## Message Passing Neural Networks (MPNN)

$\text{MPNN} \supseteq \text{GCN}$




I(•) Initialization Function — MLP

M(•) Message Function — Attention, Average, summation

U(•) Update Function — GRU, LSTM

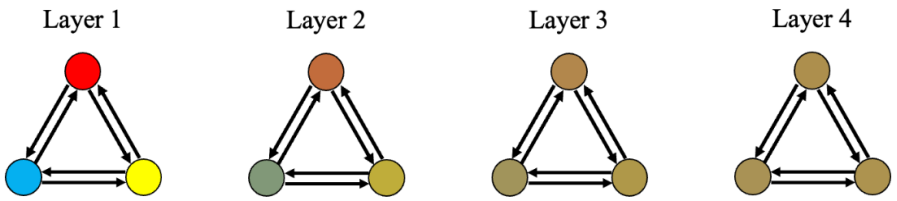
R(•) Readout Function — MLP

# An Application for Network Communication Modeling

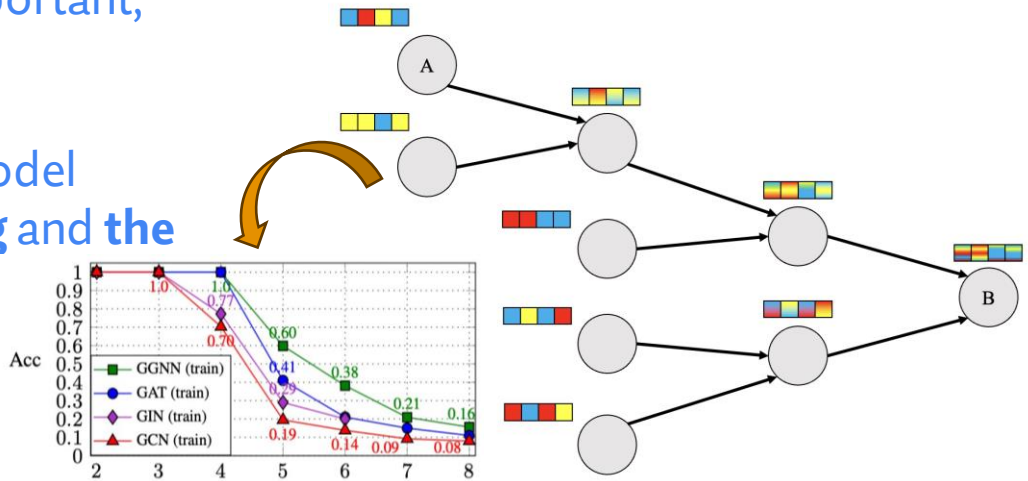
## Hyperparameters

- The size of hidden state vectors;
- The numbers of iterations.
- Setting the correct numbers for these hyperparameters is extremely important, otherwise, convergence to a good performance is unlikely;
- The main issue associated with model divergence is the **over-smoothing** and **the over-squashing:**

## Over-smoothing Problem




## Over-squashing Problem




https://minyoungg.github.io/MIT-deeplearning-blogs/2021/12/09/oversquashing-in-gnns/

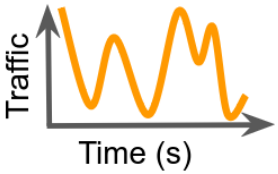
# An Application for Network Communication Modeling

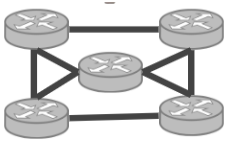
## Modeling Network with ML Models

Use the network communication inputs to mimic complex environments intended to retrieve interest metrics

Use this interest metrics to evaluate the communication network under certain conditions *(what-if analysis)*

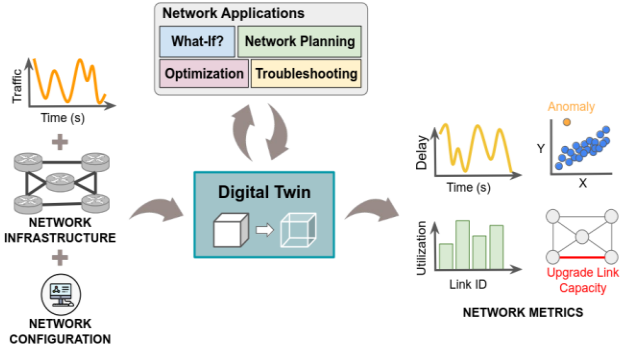
Use the evaluation results to recommend modification in the real environments

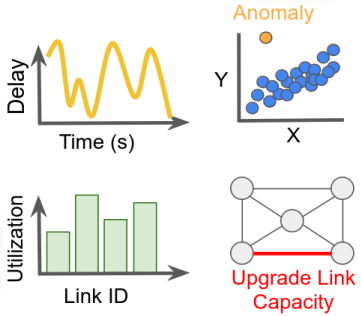


NETWORK INFRASTRUCTURE

NETWORK CONFIGURATION



# An Application for Network Communication Modeling

## Pipeline

### Dataset

**Feature Classes**

**Data Generator Input**

Network Topology L2 and L3

Routing Configuration

Traffic Configuration

Performance Matrix

### Data Generator

Define how the deep learning model will receive the data

### Input-output relation

**Network Features**

Link Capacity
Bitrate per Burst
Average Packet Size
Inter Burst Gap
⋮
Link Load

**GNN Model**

**Delay Predicted**